

# SR50 Series

## *Compact High-Resolution Rotation Stages*



**Newport®**  
Experience | Solutions

## USER'S MANUAL

# Warranty

Newport Corporation warrants this product to be free from defects in material and workmanship for a period of 1 year from the date of shipment. If found to be defective during the warranty period, the product will either be repaired or replaced at Newport's discretion.

To exercise this warranty, write or call your local Newport representative, or contact Newport headquarters in Irvine, California. You will be given prompt assistance and return instructions. Send the instrument, transportation prepaid, to the indicated service facility. Repairs will be made and the instrument returned, transportation prepaid. Repaired products are warranted for the balance of the original warranty period, or at least 90 days.

## Limitation of Warranty

This warranty does not apply to defects resulting from modification or misuse of any product or part.

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### CAUTION

Warranty does not apply to damages resulting from:

- **Incorrect usage:**
  - Load on the stage greater than maximum specified load.
  - Carriage speed higher than specified speed.
  - Improper grounding.
    - Connectors must be properly secured.
    - When the load on the stage represents an electrical risk, it must be connected to ground.
  - Excessive or improper cantilever loads.
- **Modification of the stage or any part.**



#### CAUTION

Please return equipment in the original (or equivalent) packing.

You will be responsible for damage incurred from inadequate packaging if the original packaging is not used.

This warranty is in lieu of all other warranties, expressed or implied, including any implied warranty of merchantability or fitness for a particular use. Newport Corporation shall not be liable for any indirect, special, or consequential damages.

No part of this manual may be reproduced or copied without the prior written approval of Newport Corporation.

This manual has been provided for information only and product specifications are subject to change without notice. Any changes will be reflected in future printings.

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# EC Declaration of Conformity

## SR50 Series

### EC Declaration of Conformity

following Annex II-1A  
of Directive 2006/42/EC on machinery



**The manufacturer:**

MICRO-CONTROLE Spectra-Physics,  
1 rue Jules Guesde ZI. Bois de l'Epine - BP189  
F-91006 Evry FRANCE

**Hereby declares that the machinery:**

- Description: "SR50"
- Function: Compact High-Resolution Rotation Stages
- Models: SR50CC; SR50PP.

– the technical file of which was compiled by:

Mr Dominique DEVIDAL, Quality Director,  
MICRO-CONTROLE Spectra-Physics, Zone Industrielle - B.P.29  
F-45340 Beaune La Rolande France

– complies with all the relevant provisions of the Directive 2006/42/EC on machinery.

– complies with all the relevant provisions of the Directive 2004/108/EC relating to electro-magnetic compatibility.

– was designed and built in accordance with the following harmonised standards:

- NF EN 61326-1:2006 « Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements »
- NF EN 55011:2007 Class A

– was designed and built in accordance with the following other standards:

- NF EN 61000-4-2:2009
- NF EN 61000-4-3:2006 + A2:2008
- NF EN 61000-4-4:2005
- NF EN 61000-4-6:2009

### ORIGINAL DECLARATION

Done in Beaune La Rolande on 27 January 2011  
Dominique DEVIDAL  
Quality Director

DC1-EN rev:A

# Definitions and Symbols

The following terms and symbols are used in this documentation and also appear on the product where safety-related issues occur.

**General Warning or Caution**



The exclamation symbol may appear in warning and caution tables in this document. This symbol designates an area where personal injury or damage to the equipment is possible.

**European Union CE Mark**



The presence of the CE Mark on Newport Corporation equipment means that it has been designed, tested and certified as complying with all applicable European Union (CE) regulations and recommendations.



ATTENTION

**This stage is a Class A device. In a residential environment, this device can cause radioelectric interferences. In this case, suitable measurements must be taken by the user of this device.**

**Warnings and Cautions**

The following are definitions of the Warnings, Cautions and Notes that may be used in this manual to call attention to important information regarding personal safety, safety and preservation of the equipment, or important tips.



WARNING

**Situation has the potential to cause bodily harm or death.**



CAUTION

**Situation has the potential to cause damage to property or equipment.**

NOTE

**Additional information the user or operator should consider.**

# Warnings



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**WARNING**

The rotation of objects of all types carries potential risks for operators. Ensure the protection of operators by prohibiting access to the dangerous area spotted with the opposite symbol, and by informing the personnel of the potential risks involved.

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**WARNING**

Do not use this stage when its motor is emitting smoke or is unusually hot to the touch or is emitting any unusual odor or noise or is in any other abnormal state.

Stop using the stage immediately, switch off the motor power and then disconnect the electronics power supply.

After checking that smoke is no longer being emitted contact your Newport service facility and request repairs. Never attempt to repair the stage yourself as this can be dangerous.

---

**WARNING**

Make sure that this stage is not exposed to moisture and that liquid does not get into the stage.

Nevertheless, if any liquid has entered the stage, switch off the motor power and then disconnect the electronics from power supply.

Contact your Newport service facility and request repairs.

---



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**WARNING**

Do not insert or drop objects into this stage, this may cause an electric shock, or lock the drive.

Do not use this stage if any foreign objects have entered the stage. Switch off the motor power and then disconnect the electronics power supply.

Contact your Newport service facility for repairs.

---

**WARNING**

Do not place this stage in unstable locations such as on a wobbly table or sloping surface, where it may fall or tip over and cause injury.

If this stage has been dropped or the case has been damaged, switch off the motor power and then disconnect the electronics power supply.

Contact your Newport service facility and request repairs.

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**WARNING**

Do not attempt to modify this stage; this may cause an electric shock or downgrade its performance.

---

**WARNING**

Do not exceed speed and load limitations as specified in chapter 3.3.

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**WARNING**

Do not exceed the usable depth indicated on the mounting holes (see section “Dimensions”). Longer screws can damage the mechanics or cause a short-circuit.

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# Cautions

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## CAUTION

Due to its very low profile, the SR50 stage contains pressure sensitive components. For usage at a pressure level below 0.25 bar, please contact our sales department.

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## CAUTION

Do not place this stage in a hostile environment such as X-Rays, hard UV,... or in any vacuum environment.

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## CAUTION

Do not place this stage in a location affected by dust, oil fumes or steam. This may cause an electric shock.

---

## CAUTION

Do not leave this stage in places subject to extremely high temperatures or low temperatures. This may cause an electric shock.

- Operating temperature: +10 to +35 °C.
  - Storage temperature: -10 to +40 °C (in its original packaging).
- 



## CAUTION

Do not move this stage if its motor power is on.

Make sure that the cable to the electronics is disconnected before moving the stage. Failure to do so may damage the cable and cause an electrical shock.

---

## CAUTION

Be careful that the stage is not bumped when it is being carried. This may cause it to malfunction.

---

## CAUTION

When handling this stage, always unplug the equipment from the power source for safety.

---

## CAUTION

When the carriage is in end-of-run position, it is strongly recommended not to go beyond this point by using the manual knob as this may damage the stage mechanism.

---

## CAUTION

Contact your Newport service facility to request cleaning and specification control every year.

---



# Compact High-Resolution Rotation Stages

## SR50 Series

### 1.0 Introduction

This manual provides operating instructions for the rotation stage that you have purchased in the SR50 Series:

- SR50PP
- SR50CC

#### REMARK

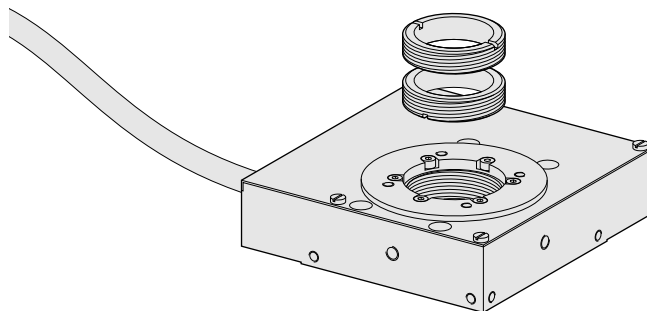
Due to its very low profile, the SR50 stage contains pressure sensitive components. For usage at a pressure level below 0.25 bar.



*SR50 rotation stage.*

#### RECOMMENDATION

We recommend you read carefully the chapter “Connection to electronics” before using the SR50 rotation stage.



*SR50 rotation stage and its 2 retaining rings.*

2.0

Description

The SR50 Series compact rotation stages are designed to rotate optical components up to 1 in. diameter, such as polarizers, wave plates, or wedges. The small size (only 77 mm width x 80 mm length x 21 mm height) allows use in confined spaces, such as laser cavities, disk texturing machines, or manufacturing areas for fiber optical components.

The lightweight, low profile design incorporates an aluminum body construction with pre-loaded ball bearings. This proprietary ball bearing design provides smooth rotation with minimal wobble. We use a precision ground worm gear that yields unvarying driving torque of the rotating carriage and eliminates variations in motor current. The motor is folded inside the stage body to give an extremely small footprint for a stage at this level of precision.

All SR50 stages feature an angular resolution of 0.001°, which may be achieved with either DC-Servo or open-loop stepper motors in half-step mode. The stages are equipped with a hardware origin allowing the stage to be returned to a reference home position to within 0.01°. SR50 stages can run in continuous rotation mode.

For optimal performance, we recommend the use of our motion controllers.

The SR50 rotation stages are equipped with a 3-meter cable for connection to our motion controllers.

2.1

Design Details

|                 |                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------|
| Base Material   | Aluminum                                                                                      |
| Bearings        | Ball bearings                                                                                 |
| Drive Mechanism | Ground worm gear                                                                              |
| Worm Gear Ratio | 1:70                                                                                          |
| Reduction Gear  | 107.143:1                                                                                     |
| Feedback        | SR50CC: Motor mounted rotary encoder, 48 pts/rev<br>SR50PP: Open-loop, no encoder             |
| Origin          | Optical                                                                                       |
| Motor           | SR50CC: DC servo motor UE16CC<br>SR50PP: 2-phase stepper motor UE16PPSC, 1 Full step = 0.002° |
| Cable (m)       | 3                                                                                             |
| MTBF            | 10,000 hours                                                                                  |



NOTE

This product complies with the RoHS directive  
(Restriction of Hazardous Substances).

## 3.0 Characteristics

### 3.1 Definitions

Specifications of our products are established in reference to ISO 230 standard part II “Determination of the position, precision and repeatability of the machine tools with CNC”.

This standard gives the definition of position uncertainty which depends on the 3 following quantities:

#### **(Absolute) Accuracy**

Difference between ideal position and real position.

#### **On-Axis Accuracy**

Difference between ideal position and real position after the compensation of linear error sources.

Linear errors include: cosine errors, inaccuracy of screw or linear scale pitch, angular deviation at the measuring point (Abbe error) and thermal expansion effects. All Newport motion electronics can compensate for linear errors.

The relation between absolute accuracy and on-axis accuracy is as follow:

$$\text{Absolute Accuracy} = \text{On-Axis Accuracy} + \text{Correction Factor} \times \text{Travel}$$

#### **Repeatability**

Ability of a system to achieve a commanded position over many attempts.

#### **Reversal Value (Hysteresis)**

Difference between actual position values obtained for a given target position when approached from opposite directions.

#### **Minimum Incremental Motion (Sensitivity)**

The smallest increment of motion a device is capable of delivering consistently and reliably.

#### **Resolution**

The smallest increment that a motion device can be commanded to move and/or detect.

#### **Concentricity**

Displacement of the geometric center of a rotation stage from the rotation axis in the plane defined by bearings.

#### **Wobble**

Tilt of rotation axis during rotation of a stage.

The testing of on-axis accuracy, repeatability, and reversal error are made systematically with our test equipment in an air-conditioned room (20 °C ±1 °C).

Each rotation stage is tested with a precision optical encoder.

A linear cycle with 21 measures on the travel and 4 cycles in each direction gives a total of 164 points.

3.2 Mechanical Specifications



|                                   |                                |
|-----------------------------------|--------------------------------|
| Travel Range (°)                  | 360 continuous                 |
| Resolution (°)                    | 0.001 <sup>(1)</sup>           |
| Minimum Incremental Motion (°)    | 0.004                          |
| Uni-directional Repeatability (°) | 0.003 typical, 0.01 guaranteed |
| Reversal Value (Hysteresis) (°)   | 0.02 typical, 0.04 guaranteed  |
| Absolute Accuracy (°)             | 0.035 typical, 0.06 guaranteed |
| Maximum Speed (°/s)               | 4 <sup>(2)</sup>               |
| Wobble (μrad)                     | 40 typical, 100 guaranteed     |

<sup>1)</sup> Equal to one half-step on SR50PP.  
<sup>2)</sup> Max. speed with SMC100PP: 1.6 °/s for SR50PP.  
See the Motion Control Metrology Primer section of the Resource Catalog for more information on typical and guaranteed specifications.



CAUTION

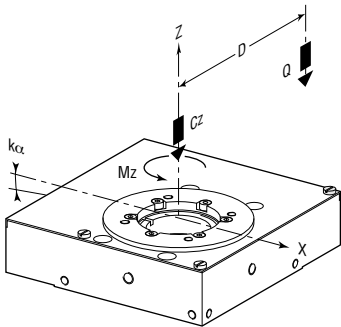
To reach specifications stated, stages must be fixed on a plane surface with a flatness of 5 μm.

3.3 Load Characteristics and Stiffness

Normal Load Capacity (Cz)

Maximum load a rotation stage can move while maintaining specifications.  
This value is given with speed and acceleration specified for each rotation stage, and with a load perpendicular to bearings.

Off-Centered Load (Q)



|     |                                         |
|-----|-----------------------------------------|
| Cz  | 30 N                                    |
| kαx | 160 μrad/Nm                             |
| Mz  | SR50CC: 0.4 Nm<br>SR50PP: 0.5 Nm        |
| Q   | Off-center load, $Q \leq Cz/(1 + D/25)$ |
| D   | Cantilever distance in mm               |
| Cz  | Normal center load capacity on bearings |
| kα  | Transversal compliance                  |

3.4 Rotation Stage Weights

The weight indicated into the below table is the value for the rotation stage with its cable installed.

|      | Weight [lb (kg)] |
|------|------------------|
| SR50 | 0.7 (0.3)        |

The weight variation between drive units is not very significant.

4.0

Drive

4.1 Stepper Drive Version

Stepper-motor-driven stages are offered in one half-step drive version: SR50PP.

Half-Step Drive

Is used for stepper motors, when 1 pulse emitted by electronic corresponds to theoretical physical motion of 1/2 of a full step of the motor.

Stepper Motor Performance Specifications

|        | Resolution<br>(°) | Speed<br>(°/s) | Motor    |
|--------|-------------------|----------------|----------|
| SR50PP | 0.001             | 4              | UE16PPSC |

NOTES

- The UE16PPSC does not use an encoder.
- The real resolution for SR50PP rotation stages is 0.000999995°.

4.2 DC-Motor Drive Version

One DC-motor-driven configuration is available: SR50CC.

DC-Motor Performance Specifications

|        | Resolution<br>(°) | Speed<br>(°/s) | Motor  |
|--------|-------------------|----------------|--------|
| SR50CC | 0.001             | 4              | UE16CC |

NOTE

The real resolution for SR50CC rotation stages is 0.000999995°.

5.0

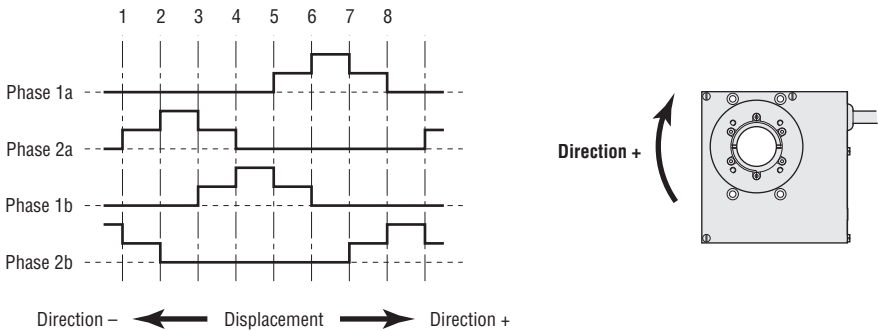
Motor

5.1 UE16PPSC Motor Characteristics

| Motor    | Angle by Step<br>(°) | Current<br>(A)                             | Resistance<br>(Ω) | Inductance<br>(mH) | Newport<br>Utilization        |
|----------|----------------------|--------------------------------------------|-------------------|--------------------|-------------------------------|
| UE16PPSC | 15                   | 0.25 <sup>(1)</sup><br>0.18 <sup>(2)</sup> | 12.5              | 5.5                | Half-Step;<br>1 phase at once |

<sup>1)</sup> 1 phase at once: 0.25 A — <sup>2)</sup> 2 phases at once: 0.18 A by phase.

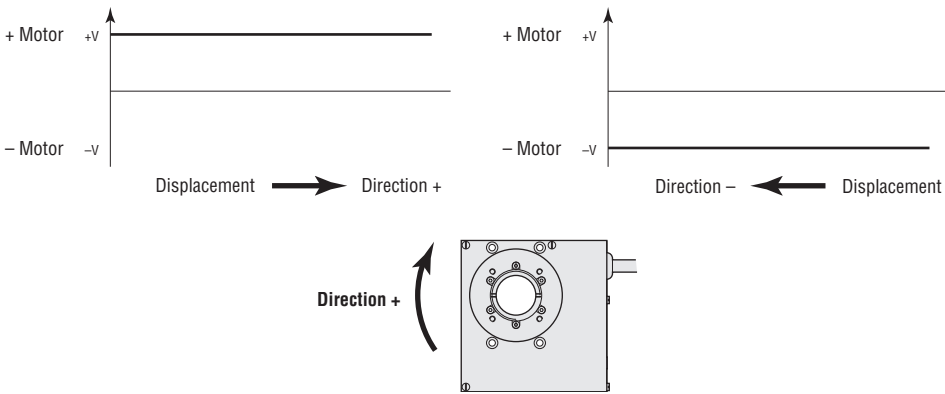
5.2 Command Signals for the Stepper Motor



5.3 UE16CC Motor Characteristics

| Motor  | Stage  | Nominal<br>Voltage (V) | Max. RMS<br>Current (A) | Max. Peak<br>Current (A) | Resistance<br>(Ω) | Inductance<br>(mH) |
|--------|--------|------------------------|-------------------------|--------------------------|-------------------|--------------------|
| UE16CC | SR50CC | 12                     | 0.075                   | 0.15                     | 117               | 0.95               |

5.4 Command Signals for the DC-Motor

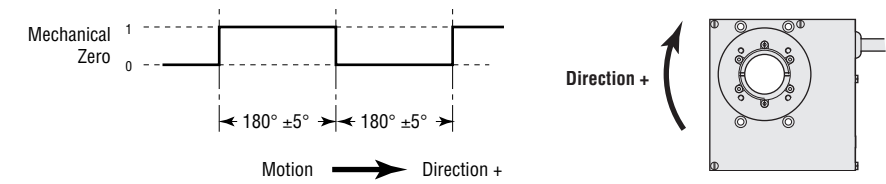


- In the above drawings, + Motor signal is referred to – Motor signal.
- 22 When the stage moves in + Direction, the + Motor voltage is higher than – Motor voltage.
- 1

When the stage moves in + Direction, the + Motor voltage is higher than – Motor voltage.
- 2

When the stage moves in – Direction, the + Motor voltage is lower than – Motor voltage.

5.5 Mechanical Zero



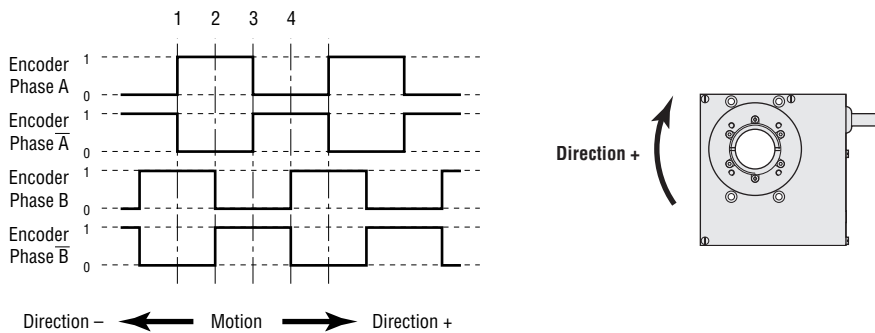
Home position signals are TTL type: 5 V ±5%, 2 mA max.  
Mechanical Zero repeatability is <0.01°.



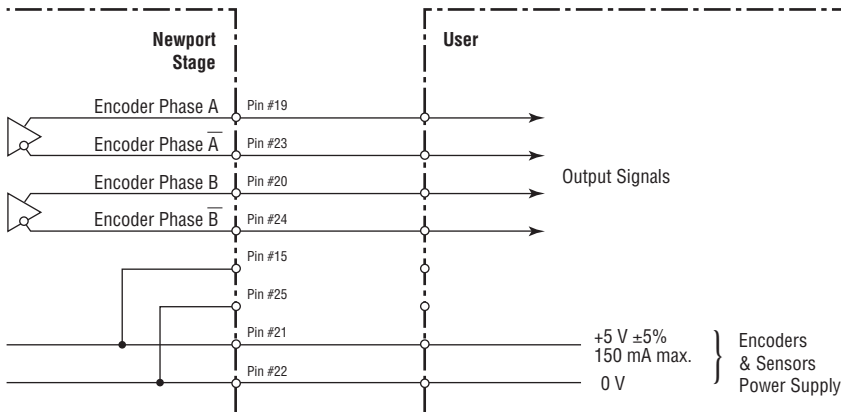
CAUTION

The “Mechanical Zero” is an active signal and should not be connected to any other source. Use appropriate TTL type receivers.

5.6 Feedback Signal Position (SR50CC Only)



The incremental sensor operates following the photoelectric measurement principle, with a disk including slides. When the sensor shaft turns, the sensor generates square signals in quadrature, sent to pins #19, #20, #23 and #24 of the 25-pin Sub-D connector.

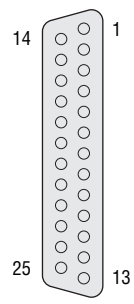




5.7 Pinouts

The 25-pin Sub-D connection for the SR50 rotation stages is given in the following table:

| UE16PPSC:<br>SR50PP |                   | UE16CC:<br>SR50CC |                     |
|---------------------|-------------------|-------------------|---------------------|
| 1                   | Phase 1a          | 1                 | N.C.                |
| 2                   | N.C.              | 2                 | N.C.                |
| 3                   | Phase 1b          | 3                 | N.C.                |
| 4                   | N.C.              | 4                 | N.C.                |
| 5                   | Phase 2a          | 5                 | + Motor             |
| 6                   | N.C.              | 6                 | + Motor             |
| 7                   | Phase 1b          | 7                 | – Motor             |
| 8                   | N.C.              | 8                 | – Motor             |
| 9                   | N.C.              | 9                 | N.C.                |
| 10                  | N.C.              | 10                | N.C.                |
| 11                  | N.C.              | 11                | N.C.                |
| 12                  | N.C.              | 12                | N.C.                |
| 13                  | Mechanical Zero   | 13                | Mechanical Zero     |
| 14                  | Shield Ground     | 14                | Shield Ground       |
| 15                  | Connected to +5 V | 15                | Connected to +5 V   |
| 16                  | 0 V logic         | 16                | 0 V logic           |
| 17                  | Do not connect    | 17                | Do not connect      |
| 18                  | Do not connect    | 18                | Do not connect      |
| 19                  | N.C.              | 19                | Encoder Phase A     |
| 20                  | N.C.              | 20                | Encoder Phase B     |
| 21                  | N.C.              | 21                | Encoder Power: +5 V |
| 22                  | N.C.              | 22                | 0 V Encoder         |
| 23                  | N.C.              | 23                | Encoder Phase /A    |
| 24                  | N.C.              | 24                | Encoder Phase /B    |
| 25                  | Connected to 0 V  | 25                | Connected to 0 V    |



## 6.0 Connection to Newport Controllers

### 6.1 Warnings on Controllers

Controllers are intended for use by qualified personnel who recognize shock hazards and are familiar with safety precautions required to avoid possible injury. Read the controller user's manual carefully before operating the instrument and pay attention to all written warnings and cautions.

---

#### WARNING

Disconnect the power plug under the following circumstances:

- If the power cord or any attached cables are frayed or damaged in any way.
- If the power plug is damaged in any way.
- If the unit is exposed to rain, excessive moisture, or liquids are spilled on the unit.
- If the unit has been dropped or the case is damaged.
- If you suspect service or repair is required.
- Whenever you clean the electronics unit.

---

#### CAUTION

To protect the unit from damage, be sure to:

- Keep all air vents free of dirt and dust.
- Keep all liquids away from the unit.
- Do not expose the unit to excessive moisture (85% humidity).
- Read this manual before using the unit for the first time.



---

#### WARNING

All attachment plug receptacles in the vicinity of this unit are to be of the grounding type and properly polarized.

Contact your electrician to check your receptacles.

---

#### WARNING

This product is equipped with a 3-wire grounding type plug.

Any interruption of the grounding connection can create an electric shock hazard.

If you are unable to insert the plug into your wall plug receptacle, contact your electrician to perform the necessary alterations to ensure that the green (green-yellow) wire is attached to earth ground.

---

#### WARNING

This product operates with voltages that can be lethal.

Pushing objects of any kind into cabinet slots or holes, or spilling any liquid on the product, may touch hazardous voltage points or short out parts.

## 6.2 Connection

On each rotation stage is represented a label which indicates its name and its serial number.



### WARNING

**Always turn the controller's power OFF before connecting to a stage.**

Stages may be connected to the rear panel motor connectors labeled "Motor..." any time prior to power-up with the supplied cable assemblies.

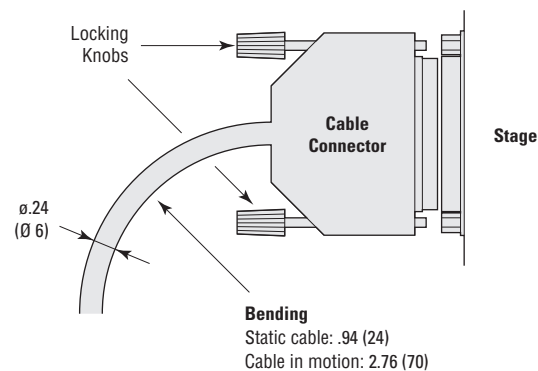
### NOTE



SR50 are ESP compatible stages. Enhanced System Performance is Newport's exclusive technology that enables Newport ESP compatible motion controllers to recognize the connected Newport ESP stage and upload the stage parameters. This ensures that the user can operate the motion system quickly and safely.

## 6.3 Cables

All our SR50 rotation stages are delivered equipped with a 3-meter cable with a 25-pin Sub-D connector so they can be directly connected to our controllers/drivers.



### WARNING

**This cable is shielded correctly. For a correct operation, make sure to lock connectors (ground continuity provided by the cable).**

For applications where the standard 3-meter cable included with your rotation stage is not adequate, Newport offers longer length cables designed to ensure the integrity of your positioning application.



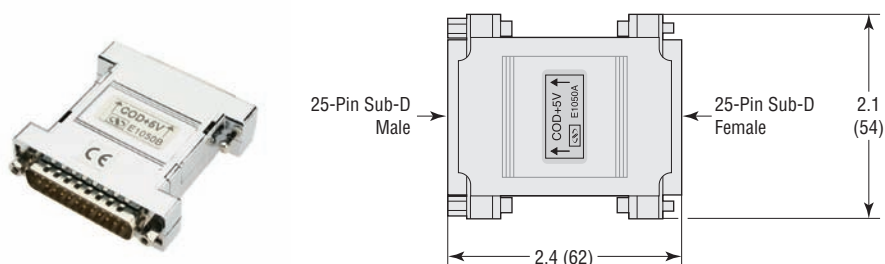
These cables are specially shielded and terminated with Newport's standard 25-pin sub-D connectors. They are available in 16.5-ft [5-m] (**MMCABLE-5**) and 23-ft [7-m] (**MMCABLE-7**).



#### WARNING

Keep the motor cables at a safe distance from other electrical cables in your environment to avoid potential cross talk.

For cable lengths in excess of 3 meters, we recommend the **MMCABLE-REG** to ensure a high quality, regulated 5 V supply to the rotation stages.



This regulator is available as an option. Please note that for best efficiency, this regulator should be attached to the stage to re-adjust the 5 volts coming from the controller through the long cable.

7.0

Connection to Non-Newport Controllers

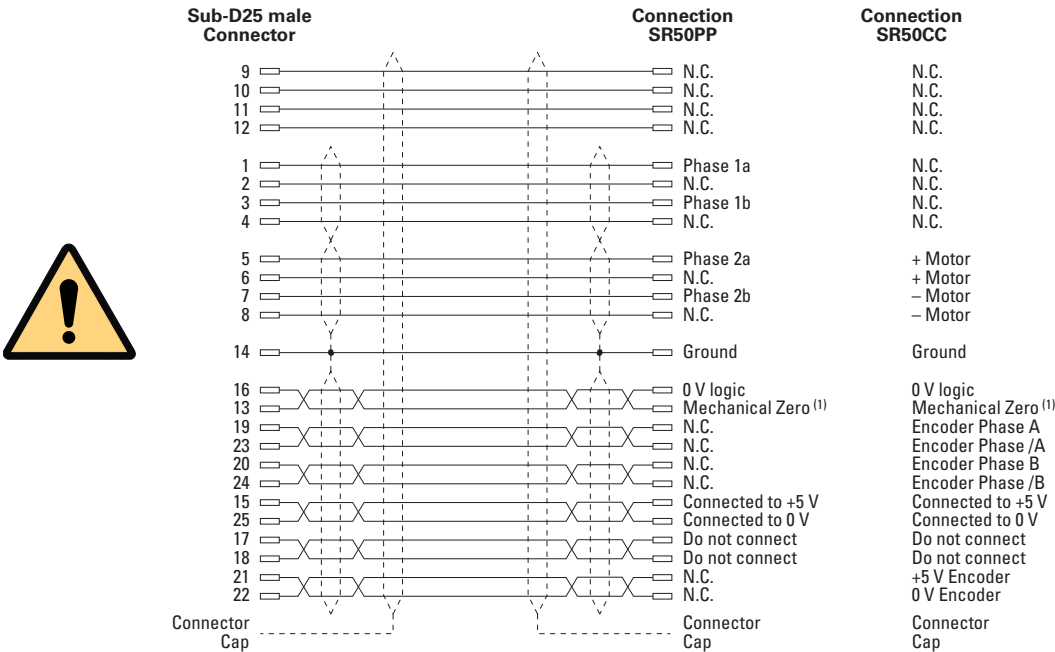
WARNING

Newport takes no responsibility for improper functioning or damage of a stage when it is used with any non- Newport controllers.

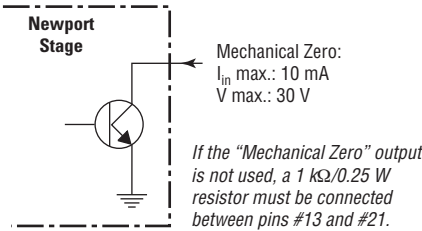
WARNING

Newport guarantees the “CE” compliance of the SR50 rotation stages only if they are used with Newport cables and controllers.

Nevertheless, the figure below indicates the recommended wiring when a SR50 rotation stage is used with non-Newport controllers.

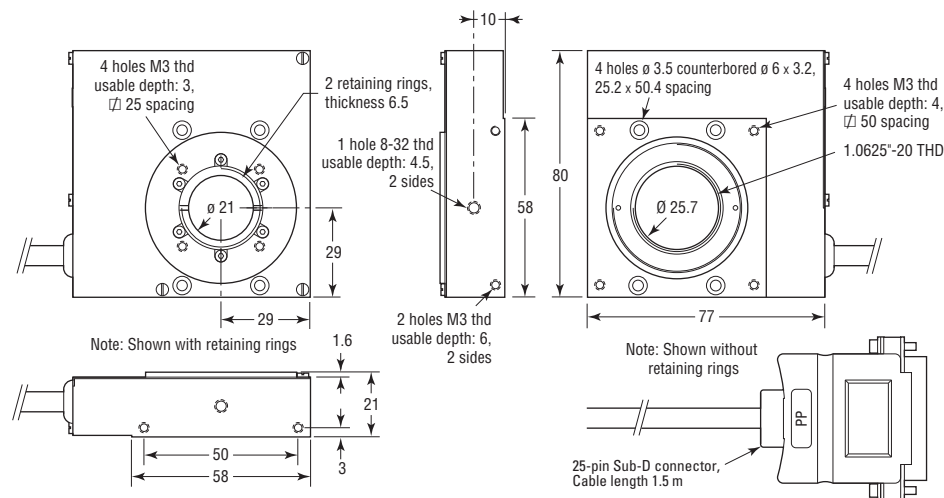


<sup>(1)</sup> The Mechanical Zero logic signal is open collector type. It supports until 30 V and 10 mA.



“Encoder” and “Index Pulse” are “differential pair” type output signals. Using these signals permits a high immunity to noise. Emission circuits generally used by Newport are 26LS31 or MC3487. Reception circuits to use are 26LS32 or MC3486.

8.0 Dimensions

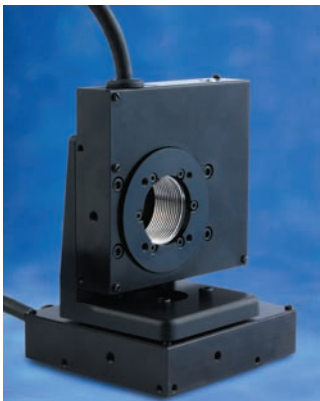
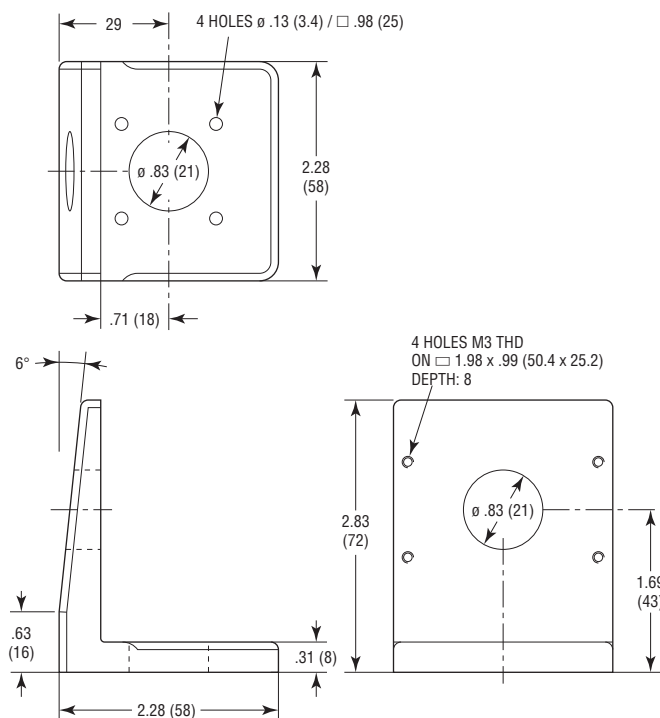


Interface

The central aperture is threaded 1.063-20 in. SR50 stages are delivered with two retaining rings with the same diameter to allow mounting of any 1" (25.4 mm) diameter optics within the aperture. 4 M3 holes allow for fixation of custom top plates.

9.0 Accessory: EQ45 Bracket

The EQ45 bracket (to order separately) can be used for vertical mounting of an SR50 rotation stage.



## 10.0 Maintenance

### RECOMMENDATION

It is recommended to contact our After Sales Service which will know to define the appropriate maintenance for your application.

### 10.1 Maintenance

The SR50 rotation stage requires no particular maintenance. Nevertheless, this is a precision mechanical device that must be kept and manipulated with precaution.

### PRECAUTIONS

The SR50 rotation stage must operate, and be stocked in a clean environment, without dust, humidity, solvents or other substances.

### RECOMMENDATION

It is recommended to return your stage to our After Sales Service after every 2000 hours of use for lubrication.

If your SR50 stage is mounted on a workstation and cannot be easily dismantled, please contact our After Sales Service for further instructions.

### 10.2 Repairing



### CAUTION

Never attempt to disassemble an element of the stage that has not been specified in this manual.

To disassemble a non specified element can cause a malfunction of the stage.

If you observe a malfunction in your stage, please immediately contact us to make arrangements for a repair.



### CAUTION

All disassembly attempts or repair of rotation stage without authorization will void your warranty.

### 10.3 Calibration



### CAUTION

It is recommended to return your rotation stage to Newport once a year for a recalibration to its original specifications.

# Service Form

## Your Local Representative

Tel.: \_\_\_\_\_

Fax: \_\_\_\_\_

Name: \_\_\_\_\_

Return authorization #: \_\_\_\_\_

(Please obtain prior to return of item)

Company: \_\_\_\_\_

Address: \_\_\_\_\_

Date: \_\_\_\_\_

Country: \_\_\_\_\_

Phone Number: \_\_\_\_\_

P.O. Number:

P.O. Number: Fax Number:

**Item(s) Being Returned:**

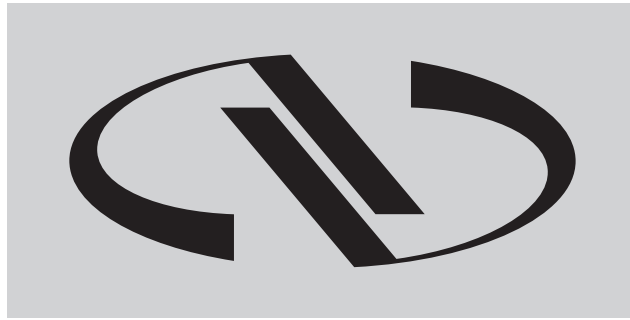
Model #: \_\_\_\_\_

Serial #: \_\_\_\_\_

Description: \_\_\_\_\_

Reasons of return of goods (please list any specific problems): \_\_\_\_\_





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